

Chapter 14: Orthogonal Vectors Worksheet

1. Are each of the following statements true or false?

- (a) If $\mathbf{x}$ and $\mathbf{y}$ are orthogonal, so are $\mathbf{x}$ and $\mathbf{x} + \mathbf{y}$.
- (b) If $\mathbf{u}$ is orthogonal to both $\mathbf{x}$ and $\mathbf{y}$, then $\mathbf{u}$ is orthogonal to $\mathbf{x} + \mathbf{y}$.
- (c) If $\mathbf{u}$ is parallel to $\mathbf{v}$, then $\text{proj}_{\mathbf{v}}\mathbf{u} = \mathbf{v}$.

2. Find a, b such that $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is perpendicular/orthogonal to $\begin{bmatrix} a \\ 0 \\ \frac{b}{3} \end{bmatrix}$, and $\begin{bmatrix} a \\ 0 \\ \frac{b}{3} \end{bmatrix}$ is a unit vector.

3. Let $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix}$ be a vector in $\mathbb{R}^3$. Describe the set of vectors which are orthogonal to $\mathbf{u}$ as a parametrized set.

4. Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are pair-wise orthogonal unit vectors in $\mathbb{R}^3$ (note: this is called an orthonormal set). Let $\mathbf{u} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3$ be a linear combination of these vectors.

(a) Compute $\mathbf{u} \cdot \mathbf{v}_1, \mathbf{u} \cdot \mathbf{v}_2, \mathbf{u} \cdot \mathbf{v}_3$ using the above information.

(b) Show that $\mathbf{w}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \mathbf{w}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{w}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is a set of pair-wise orthogonal unit vectors.

(c) Using your work from part (a), find the coefficients which make $\mathbf{x} = \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix}$ a linear combination of $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$.